

INTERNSHIP FINAL PROJECT REPORT

ON

**DESIGN AND VERIFICATION
OF 32-BIT ALU**

Submitted by

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In partial fulfillment of the requirements for the
Internship Final Project

Department of Electronics and Communication Engineering

Academic Year 2025–2026

CERTIFICATE

This is to certify that the project work entitled “DESIGN AND VERIFICATION OF 32-BIT ALU USING VERILOG HDL AND VIVADO” submitted by ASHRITH PATEL is a bonafide record of work carried out as part of the Internship Final Project during the academic year 2025–2026.

The project has been completed successfully under the prescribed guidelines and demonstrates the design, implementation, simulation, verification, RTL analysis, and synthesis of a 32-bit Arithmetic Logic Unit using Verilog Hardware Description Language and Vivado Design Suite.

The work presented in this report is original and has not been submitted previously for the award of any degree, diploma, or certification. The project reflects the candidate’s understanding of digital system design, FPGA design flow, simulation techniques, and hardware verification methodologies.

This report has been approved as a partial fulfillment of the requirements for the successful completion of the Internship Final Project in the field of Electronics and Communication Engineering.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported and guided me during the completion of this internship project on “Design and Verification of 32-bit ALU Using Verilog HDL and Vivado”.

I gained valuable knowledge and practical experience in digital system design, Verilog HDL coding, simulation, RTL analysis, and synthesis using Vivado Design Suite throughout this project work.

I would also like to thank the internship organization and all individuals who provided support, encouragement, and technical guidance during the successful completion of this project.

Finally, I would like to thank my family and friends for their continuous motivation and support throughout the project duration.

ABSTRACT

The Arithmetic Logic Unit (ALU) is one of the fundamental building blocks of modern digital systems and processor architectures. It is responsible for performing arithmetic and logical operations that are essential for computation and data processing in computer systems. The performance and efficiency of a processor largely depend on the functionality of the ALU.

This project presents the design and verification of a 32-bit Arithmetic Logic Unit using Verilog Hardware Description Language (HDL) and Vivado Design Suite. The designed ALU is capable of executing multiple arithmetic and logical operations such as addition, subtraction, bitwise AND, OR, XOR, NOT, left shift, and right shift operations based on the applied control signals.

The ALU design was implemented using behavioral modeling techniques in Verilog HDL. A comprehensive testbench was developed to verify the correctness and functionality of the design through behavioral simulation using Vivado Simulator. The simulation waveforms confirmed the successful execution of all ALU operations.

RTL analysis and synthesis were carried out successfully to evaluate the hardware implementation and FPGA resource utilization. The synthesis reports provided details regarding LUT utilization, carry logic usage, and I/O resource consumption. Timing analysis was also performed to verify the operational reliability of the design.

This project provides practical exposure to digital system design, RTL coding, functional verification, simulation, synthesis flow, and FPGA-based implementation methodologies used in VLSI and embedded system applications. The developed 32-bit ALU can further be extended for advanced processor architectures and high-performance computing applications.

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CHAPTER 1

INTRODUCTION

An Arithmetic Logic Unit (ALU) is one of the most essential components of a digital computer system and processor architecture. It performs arithmetic and logical operations required for computational tasks and data processing in modern digital systems. The ALU acts as the computational core of the Central Processing Unit (CPU) and plays a major role in executing instructions and processing binary data.

In modern processors, the ALU is responsible for performing arithmetic operations such as addition and subtraction, along with logical operations including AND, OR, XOR, and NOT. It also supports shift operations which are important in digital signal processing, arithmetic calculations, and data manipulation tasks.

With the rapid growth of digital electronics and embedded systems, efficient ALU design has become an important aspect of VLSI system design. The implementation of ALUs using Hardware Description Languages (HDLs) allows designers to model, simulate, verify, and synthesize digital circuits effectively before hardware implementation.

This project focuses on the design and verification of a 32-bit Arithmetic Logic Unit using Verilog HDL and Vivado Design Suite. The ALU was designed using behavioral modeling techniques in Verilog and supports multiple arithmetic and logical operations based on control signals.

The functionality of the ALU was verified through simulation using Vivado Simulator. RTL analysis and synthesis were performed successfully to analyze hardware implementation and FPGA resource utilization. The project also provides practical exposure to digital design flow, RTL coding, simulation methodologies, synthesis flow, and FPGA-based system implementation techniques widely used in VLSI and embedded system applications.

1.1 Overview of ALU

The Arithmetic Logic Unit is a combinational digital circuit that performs mathematical and logical operations on binary data inputs. It receives operands from processor registers and performs operations according to control signals generated by the control unit.

The ALU is capable of handling arithmetic operations such as addition and subtraction, logical operations such as AND, OR, XOR, and NOT, and shift operations including left shift and right shift. The output generated by the ALU is then stored back into registers or forwarded for further processing within the processor architecture.

A 32-bit ALU processes 32-bit binary data, making it suitable for modern computing applications and processor systems. The use of 32-bit architecture improves computational capability and data handling efficiency compared to lower-bit ALU architectures.

1.2 Importance of ALU

The ALU is considered the heart of every processor because all arithmetic calculations and logical decision-making operations are performed within it. Every digital computing system depends on the ALU for executing instructions and processing numerical and logical data efficiently.

The performance of a processor is greatly influenced by the speed and efficiency of the ALU design. Modern applications such as embedded systems, signal processing, communication systems, and computing devices require high-speed and reliable ALU architectures for efficient operation.

1.3 Verilog HDL

Verilog Hardware Description Language (HDL) is widely used for modeling, simulation, and implementation of digital electronic systems. It provides a hardware-oriented programming approach that allows designers to describe digital circuits at various levels of abstraction.

Verilog HDL supports behavioral, dataflow, and structural modeling techniques, making it suitable for designing combinational and sequential digital circuits. In this project, behavioral modeling was used to implement the 32-bit ALU operations efficiently.

Verilog HDL also allows simulation and verification of the design before actual hardware implementation. This helps designers identify and correct errors during the development phase, reducing hardware complexity and design cost.

CHAPTER 2

OBJECTIVES

2.1 Project Objectives

The main objective of this project is to design and verify a 32-bit Arithmetic Logic Unit (ALU) using Verilog Hardware Description Language and Vivado Design Suite. The project aims to understand the implementation of arithmetic and logical operations using digital hardware design methodologies.

The project focuses on developing a functional ALU capable of performing arithmetic operations such as addition and subtraction, along with logical operations including AND, OR, XOR, and NOT. Shift operations such as left shift and right shift are also implemented as part of the ALU functionality.

Another important objective of the project is to perform functional verification using a Verilog testbench and behavioral simulation techniques. RTL analysis and synthesis are carried out to study hardware implementation and FPGA resource utilization.

The project also aims to provide practical exposure to digital system design, RTL coding, simulation, synthesis flow, and FPGA-based implementation methodologies used in VLSI and embedded system applications.

2.2 Scope of the Project

The scope of this project includes the design, implementation, simulation, and verification of a 32-bit Arithmetic Logic Unit using Verilog HDL. The ALU is designed as a combinational digital circuit capable of performing multiple arithmetic and logical operations based on control signals.

The project primarily focuses on behavioral modeling techniques and simulation-based verification using Vivado Simulator. RTL analysis and synthesis are performed to analyze hardware implementation and FPGA resource utilization.

The developed ALU architecture can further be extended to support advanced arithmetic operations such as multiplication and division. The design can also be integrated into processor architectures, embedded systems, and FPGA-based digital systems for practical hardware implementation.

This project serves as a foundation for understanding advanced VLSI design concepts, processor design, FPGA implementation, and digital hardware verification techniques.

CHAPTER 3

LITERATURE SURVEY

3.1 Digital System Design

Digital system design is an important field in electronics and computer engineering that deals with the design and implementation of digital circuits and systems. Modern digital systems are widely used in computers, communication systems, embedded devices, and consumer electronics.

Digital circuits are generally categorized into combinational and sequential circuits. Combinational circuits generate outputs based only on present input values, whereas sequential circuits depend on both present inputs and previous states. Arithmetic Logic Units are examples of combinational digital circuits.

The advancement of Hardware Description Languages such as Verilog HDL has simplified the design and verification process of digital systems. HDL-based design enables simulation, synthesis, testing, and hardware implementation before physical realization of the circuit.

3.2 Arithmetic Logic Unit

The Arithmetic Logic Unit is one of the core components of a processor responsible for performing arithmetic and logical operations. ALUs are widely used in microprocessors, microcontrollers, digital signal processors, and embedded systems.

Different ALU architectures have been developed to improve computational speed, reduce delay, and optimize hardware resource utilization. The complexity and performance of an ALU depend on the number of supported operations and the bit-width of the architecture.

A 32-bit ALU is commonly used in modern digital systems because it provides improved computational capability and efficient handling of larger binary data. Arithmetic operations such as addition and subtraction and logical operations such as AND, OR, XOR, and NOT are fundamental operations performed by ALUs.

3.3 FPGA and Verilog HDL

Field Programmable Gate Arrays (FPGAs) are programmable semiconductor devices used for implementing digital hardware systems. FPGAs provide flexibility, reconfigurability, and faster development cycles compared to traditional hardware implementation methods.

Verilog Hardware Description Language is extensively used for FPGA-based digital design because it allows designers to model hardware circuits efficiently. Verilog supports behavioral, dataflow, and structural modeling techniques for implementing digital systems.

Vivado Design Suite is a widely used FPGA development tool that supports design entry, simulation, RTL analysis, synthesis, and implementation of Verilog-based digital systems. In this project, Verilog HDL and Vivado were used for designing, simulating, verifying, and synthesizing the 32-bit ALU successfully.

CHAPTER 4

SYSTEM DESIGN

4.1 Block Diagram of 32-bit ALU

The system design of the 32-bit Arithmetic Logic Unit consists of two 32-bit input operands, a 4-bit control signal, and output signals. The ALU performs arithmetic and logical operations depending on the control signal provided to the design.

The two 32-bit inputs are represented as A and B, while the control signal ALU_Sel determines the operation to be executed. The output generated by the ALU is represented by ALU_Out, and a Zero flag is used to indicate whether the output result is zero.

The ALU architecture is designed using combinational logic and behavioral Verilog modeling techniques. The overall system design allows efficient execution of arithmetic and logical operations required for digital processing applications.

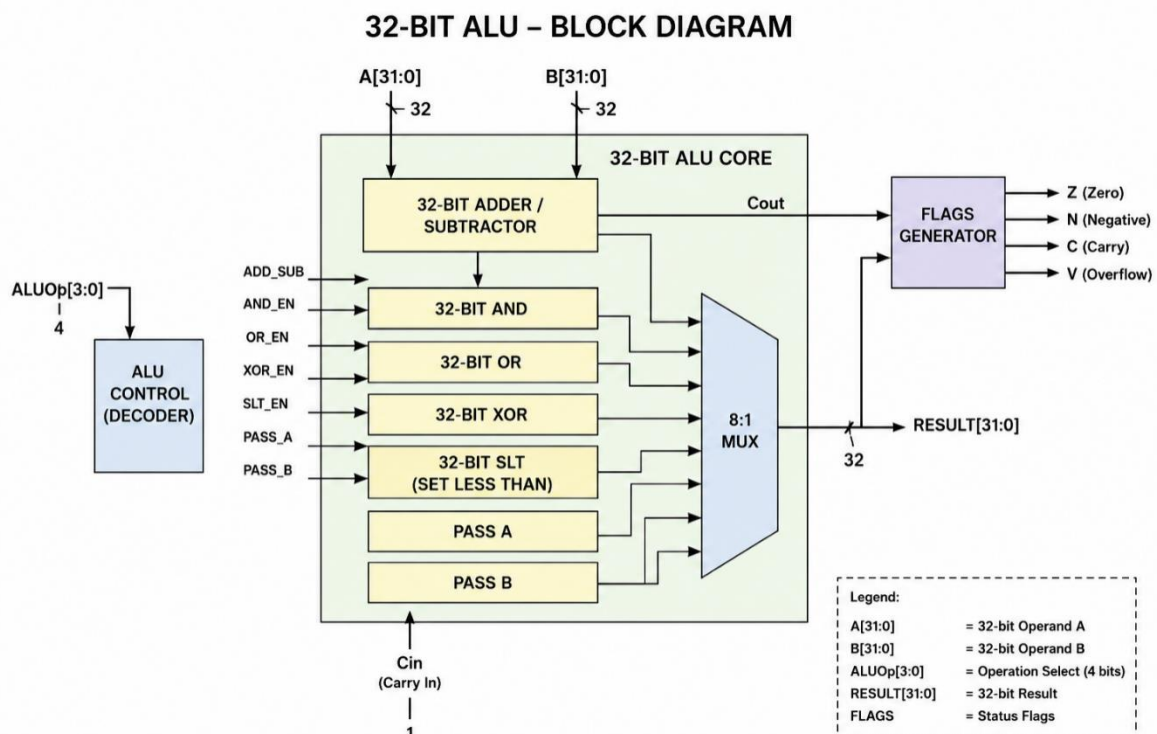


Figure 4.1: Block Diagram of 32-bit ALU

ALU_Sel	Operation
0000	Addition
0001	Subtraction
0010	AND
0011	OR
0100	XOR
0101	NOT
0110	Left Shift
0111	Right Shift

Table 4.1: ALU Operations and Control Signals

4.2 ALU Functional Operations

The designed 32-bit ALU supports multiple arithmetic and logical operations based on the control signal input. Arithmetic operations include addition and subtraction, while logical operations include AND, OR, XOR, and NOT operations. The ALU also performs left shift and right shift operations.

Each operation is selected using a unique 4-bit control signal. The ALU processes the input operands and generates the corresponding output result according to the selected operation. The implementation was carried out using case statements in behavioral Verilog HDL.

The supported operations improve the functionality and flexibility of the ALU architecture, making it suitable for digital processing and embedded system applications.

4.3 Input and Output Description

The input and output ports of the 32-bit ALU are designed to support arithmetic and logical processing efficiently. Inputs A and B are 32-bit operands provided to the ALU for performing computations. The ALU_Sel signal acts as a control input for selecting the required operation.

The ALU_Out signal represents the 32-bit output generated after execution of the selected operation. The Zero flag becomes active whenever the output result is equal to zero. This flag is useful in processor architectures and conditional operations.

CHAPTER 5

METHODOLOGY

5.1 Design Flow

The design methodology followed in this project includes digital design entry, Verilog HDL coding, simulation, RTL analysis, synthesis, and hardware resource utilization analysis using Vivado Design Suite.

Initially, the functionality and architecture of the 32-bit ALU were studied in detail. The required arithmetic and logical operations were identified and implemented using behavioral Verilog HDL coding techniques.

After completion of the Verilog module, a testbench was developed to apply various input combinations and verify the correctness of ALU operations through simulation. Behavioral simulation was carried out using Vivado Simulator to validate the functionality of the design.

RTL analysis and synthesis were then performed to analyze the hardware implementation and FPGA resource utilization. The synthesis reports provided details regarding LUT utilization, I/O usage, and timing analysis of the designed ALU.

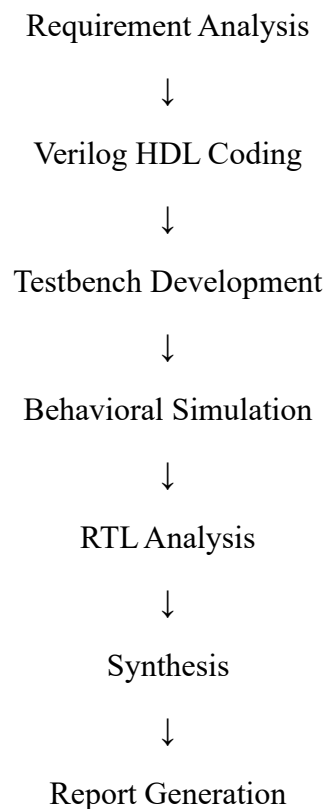


Figure 5.1: Design Methodology Flow

5.2 Verilog Modeling

Behavioral modeling technique was used to design the 32-bit ALU in Verilog HDL. In behavioral modeling, the functionality of the digital circuit is described using procedural statements and control structures.

The ALU operations were implemented using case statements based on the control signal input. Arithmetic operations such as addition and subtraction and logical operations such as AND, OR, XOR, and NOT were implemented efficiently using Verilog operators.

Behavioral modeling simplifies the implementation process and improves readability and maintainability of the design code. It also enables easy verification and modification of ALU functionality.

5.3 Functional Simulation

Functional simulation was performed to verify the correctness of the 32-bit ALU design. A Verilog testbench was created to apply different input combinations and control signals to the ALU module.

Vivado Simulator was used to execute the behavioral simulation and generate waveform outputs. The simulation results confirmed that the ALU performed all arithmetic and logical operations correctly according to the applied control signals.

Simulation plays an important role in digital system design because it helps identify design errors before hardware implementation. It also ensures proper functionality and reliability of the system.

5.4 RTL Analysis

Register Transfer Level (RTL) analysis was performed using Vivado Design Suite after successful behavioral simulation. RTL analysis provides a graphical representation of the hardware architecture generated from the Verilog HDL code.

The RTL schematic of the designed ALU shows the logical connections, combinational blocks, and data flow within the system. RTL analysis helps verify whether the synthesized hardware architecture matches the intended digital design.

The generated RTL schematic confirmed the correct implementation of arithmetic and logical operations in the designed 32-bit ALU.

5.5 Synthesis Process

Synthesis is the process of converting Verilog HDL code into hardware-level gate implementation suitable for FPGA realization. In this project, Vivado synthesis tool was used to synthesize the designed 32-bit ALU.

The synthesis process generated reports related to LUT utilization, carry logic usage, I/O resource usage, and timing analysis. These reports help evaluate hardware complexity and resource consumption of the design.

Successful synthesis of the ALU confirmed that the Verilog design was functionally correct and suitable for FPGA-based implementation.

CHAPTER 6

VERILOG IMPLEMENTATION

6.1 ALU Verilog Code

The 32-bit Arithmetic Logic Unit was implemented using behavioral Verilog HDL modeling techniques. The ALU supports arithmetic operations such as addition and subtraction, logical operations such as AND, OR, XOR, and NOT, and shift operations including left shift and right shift.

The operation to be executed is selected using a 4-bit control signal. Case statements were used in the Verilog code to perform the required operations efficiently.

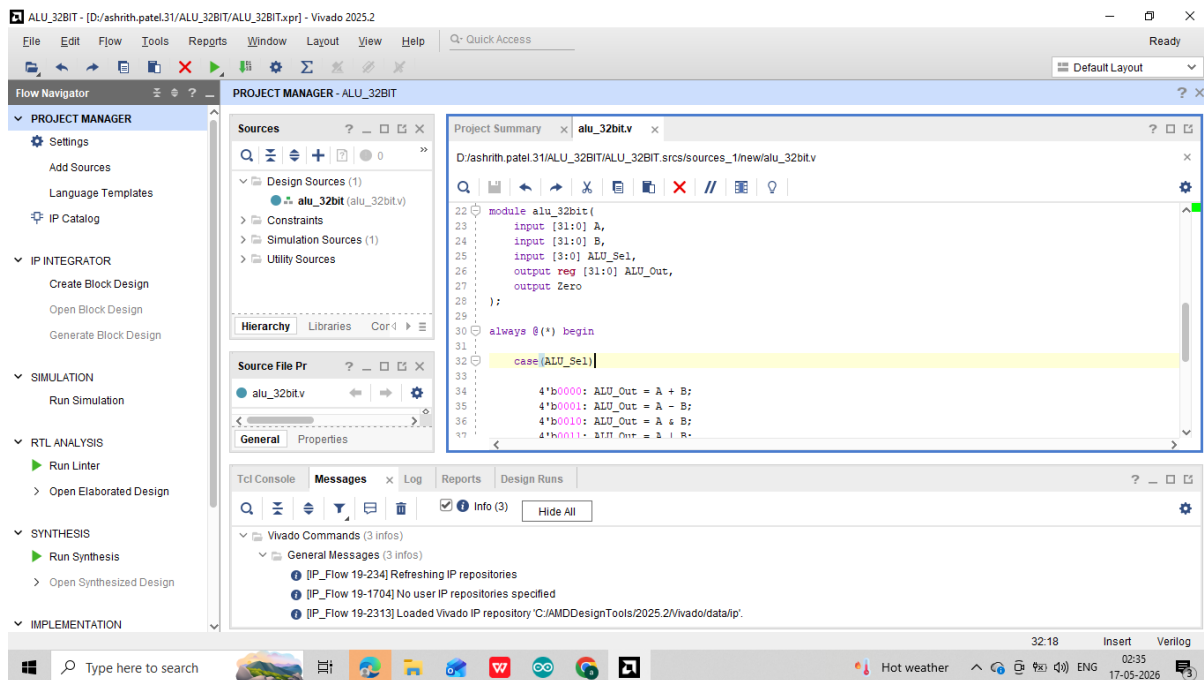
```
module alu_32bit(  
    input [31:0] A,  
    input [31:0] B,  
    input [3:0] opcode,  
    output reg [31:0] Result,  
    output reg Carry,  
    output reg Zero );  
    always @(*) begin  
        case(opcode)  
            4'b0000: Result = A + B;  
            4'b0001: Result = A - B;  
            4'b0010: Result = A & B;  
            4'b0011: Result = A | B;  
            4'b0100: Result = A ^ B;  
            4'b0101: Result = ~A;  
            4'b0110: Result = A << 1;  
            4'b0111: Result = A >> 1;  
            default: Result = 32'b0;  
        endcase
```

```
Zero = (Result == 0);
```

```
end
```

```
endmodule
```

Program 6.1: Verilog Code for 32-bit ALU



Program 6.2: Verilog Code for 32-bit ALU in Vivado sSimulator

6.2 Description of ALU Operations

The designed ALU performs multiple arithmetic and logical operations based on the value of the control signal ALU_Sel. Each operation is assigned a unique control code which determines the functionality executed by the ALU.

Addition and subtraction operations are used for arithmetic computation, while AND, OR, XOR, and NOT operations are used for logical data processing. Shift operations are included for bit manipulation and digital processing applications.

The use of behavioral Verilog modeling and case statements simplifies implementation and improves readability of the ALU design.

ALU_Sel	Operation
0000	Addition
0001	Subtraction
0010	AND
0011	OR
0100	XOR
0101	NOT
0110	Left Shift
0111	Right Shift

Table 6.1: ALU Operations and Control Signals

6.3 Code Explanation

The Verilog module consists of input operands A and B, control signal ALU_Sel, output signal ALU_Out, and Zero flag. The always block is used to execute operations whenever input values or control signals change.

Case statements are used to select arithmetic and logical operations based on the ALU_Sel input. Arithmetic operators and logical operators available in Verilog HDL were used for implementing the required operations.

The Zero flag is generated by comparing the output result with zero. If the output becomes zero, the Zero flag is activated. This feature is commonly used in processor architectures for conditional operations and branching instructions.

CHAPTER 7

TESTBENCH

7.1 Testbench Code

A Verilog testbench was developed to verify the functionality and correctness of the designed 32-bit ALU. The testbench applies different combinations of input operands and control signals to the ALU module and observes the generated outputs.

The testbench plays an important role in validating arithmetic and logical operations before hardware implementation. Different ALU operations such as addition, subtraction, AND, OR, XOR, NOT, left shift, and right shift were tested successfully using simulation.

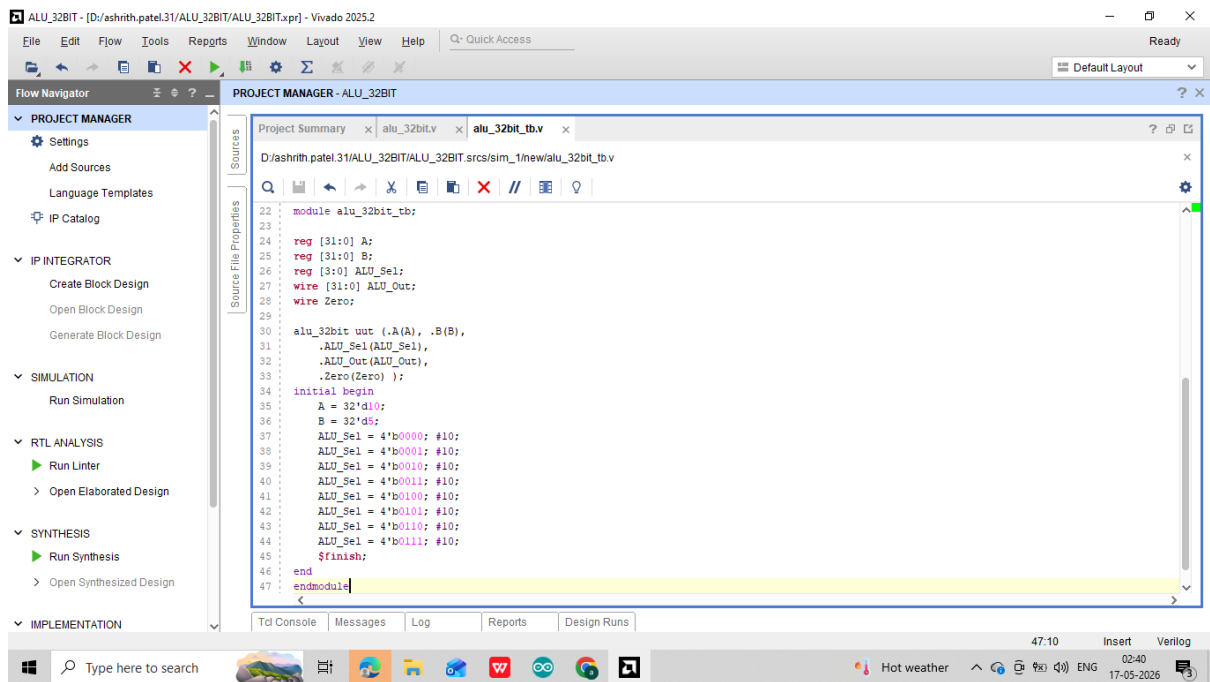
```
module alu_tb;
reg [31:0] A, B;
reg [3:0] opcode;
wire [31:0] Result;
wire Carry, Zero;

alu_32bit uut( .A(A), .B(B), .opcode(opcode), .Result(Result) .Carry(Carry), .Zero(Zero)
);

initial begin
A = 32'd10;
B = 32'd5;
opcode = 4'b0000;
#10; opcode = 4'b0001;
#10; opcode = 4'b0010;
#10; opcode = 4'b0011;
#10;
$finish;
end

endmodule
```

Program 7.1: Verilog Testbench for 32-bit ALU



Program 7.1: Verilog Testbench for 32-bit ALU in Vivado Simulator

7.2 Verification Process

The verification process was carried out using Vivado Simulator. Different test cases were applied to the ALU inputs through the testbench module, and waveform outputs were analyzed to verify functionality.

The simulation results confirmed that all arithmetic and logical operations were executed correctly according to the control signals. Functional verification ensured the reliability and correctness of the ALU design before synthesis and hardware implementation.

CHAPTER 8

SIMULATION RESULTS

8.1 Behavioral Simulation

Behavioral simulation was performed using Vivado Simulator to verify the functionality of the 32-bit ALU. The simulation process generated waveform outputs for different arithmetic and logical operations.

The testbench provided multiple input combinations and control signals to validate the performance of the ALU. The simulation results confirmed successful execution of addition, subtraction, AND, OR, XOR, NOT, left shift, and right shift operations.

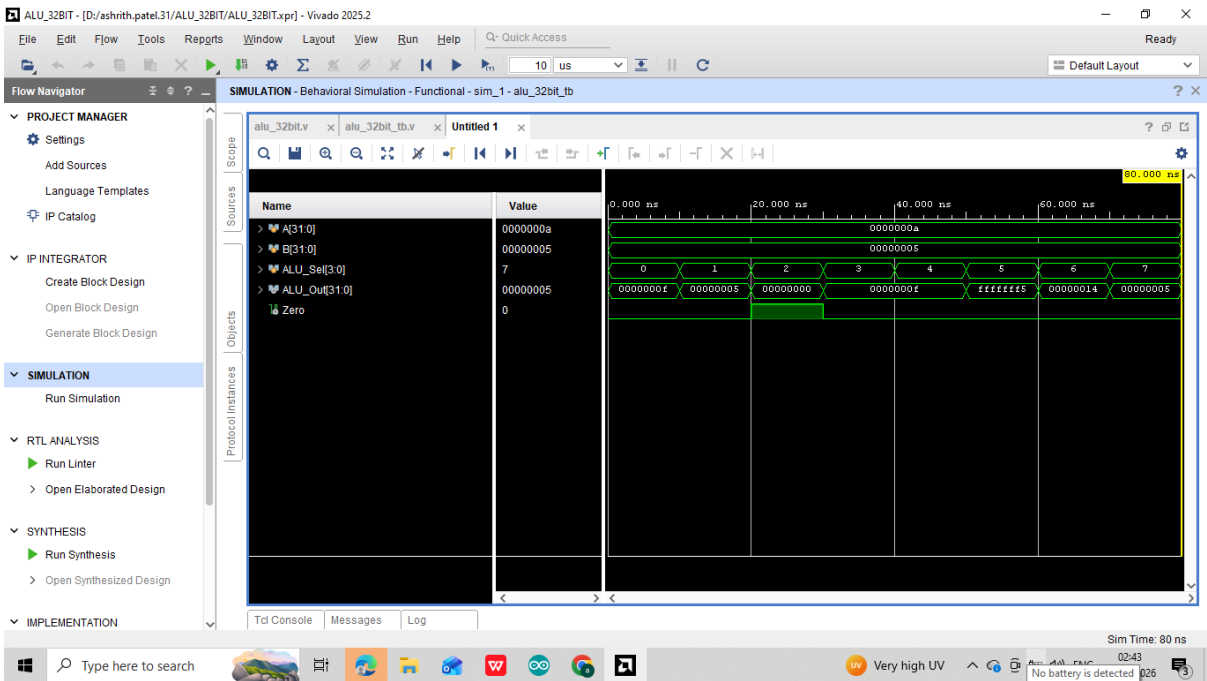


Figure 8.1: Behavioral Simulation Waveform

8.2 Waveform Analysis

The generated simulation waveforms were analyzed to verify the correctness of the ALU outputs. Each control signal produced the expected arithmetic or logical operation according to the designed functionality.

The waveform analysis confirmed that the ALU generated accurate outputs for all input test cases. The timing behavior and signal transitions observed in the simulation validated the correctness and stability of the design.

CHAPTER 9

RTL SCHEMATIC

9.1 Elaborated RTL Design

RTL analysis was performed after successful simulation using Vivado Design Suite. The Register Transfer Level schematic provides a graphical representation of the hardware architecture generated from the Verilog HDL code.

The RTL schematic shows the logical connections and internal architecture of the designed 32-bit ALU. It represents arithmetic and logical operation blocks along with data flow between input and output ports.

RTL analysis helps designers verify whether the generated hardware architecture matches the intended design functionality before synthesis and implementation.

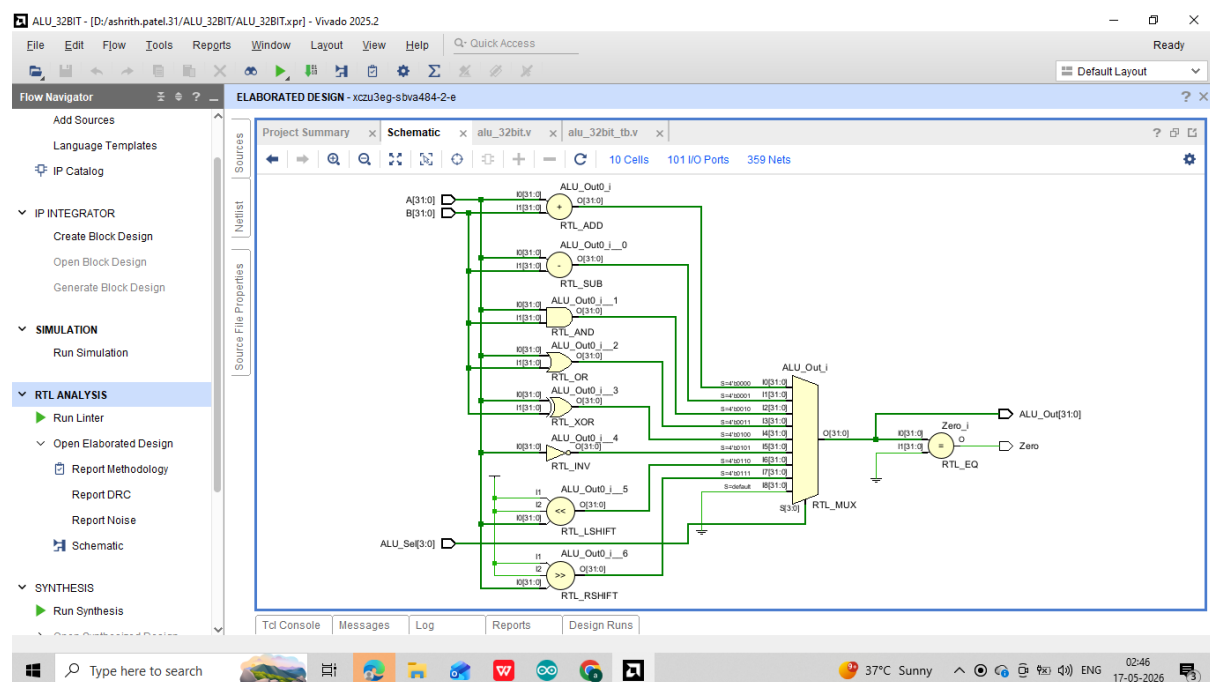


Figure 9.1: RTL Schematic of 32-bit ALU

9.2 RTL Schematic Explanation

The RTL schematic generated by Vivado confirmed the successful implementation of arithmetic and logical operations in the designed ALU architecture. The schematic displayed combinational logic structures, multiplexing operations, and signal routing within the design.

The RTL representation verified that the Verilog HDL code was synthesized correctly into hardware-level architecture suitable for FPGA implementation.

CHAPTER 10

SYNTHESIS REPORT

10.1 Synthesized Design

After successful RTL analysis, synthesis was performed using Vivado Design Suite. Synthesis converts the Verilog HDL design into gate-level hardware representation suitable for FPGA implementation.

The synthesized design confirmed successful hardware realization of the 32-bit ALU architecture. Vivado generated synthesis reports related to LUT utilization, I/O usage, and timing analysis.

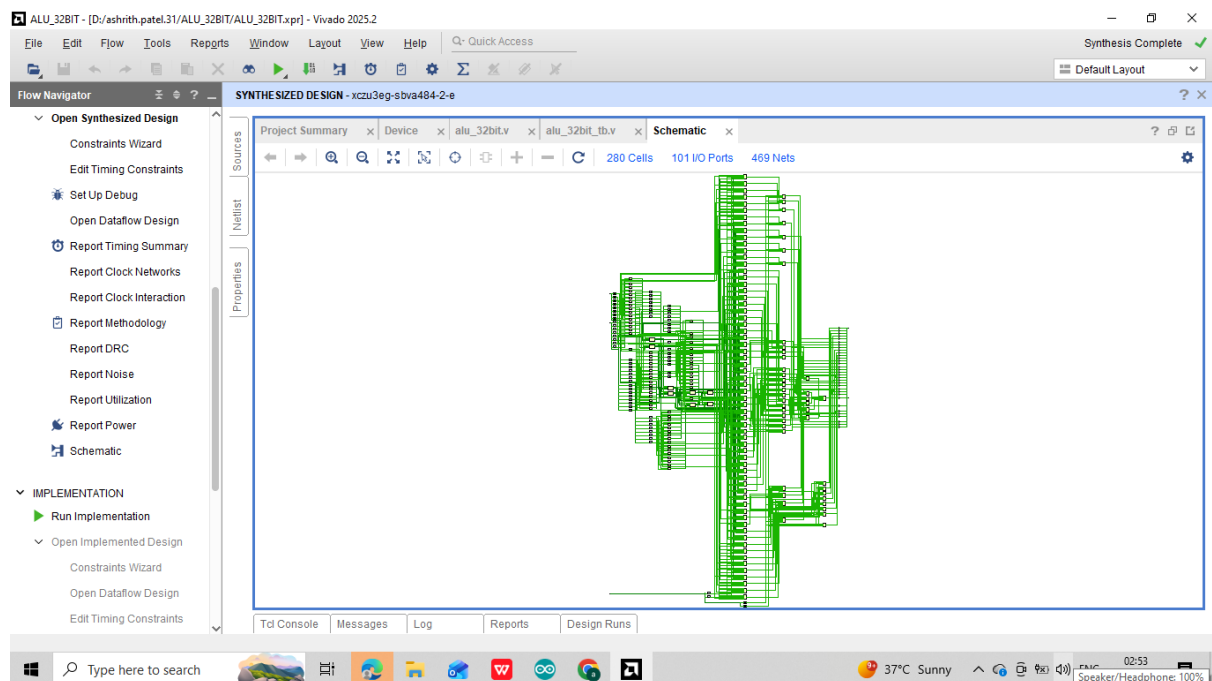


Figure 10.1: Synthesized Design of 32-bit ALU

10.2 Utilization Report

The utilization report generated during synthesis provides details regarding FPGA resource consumption by the ALU design. The report includes information about LUT usage, flip-flop utilization, carry logic usage, and I/O resources.

The synthesis results showed that the designed ALU utilized FPGA resources efficiently while maintaining proper functionality and hardware performance.

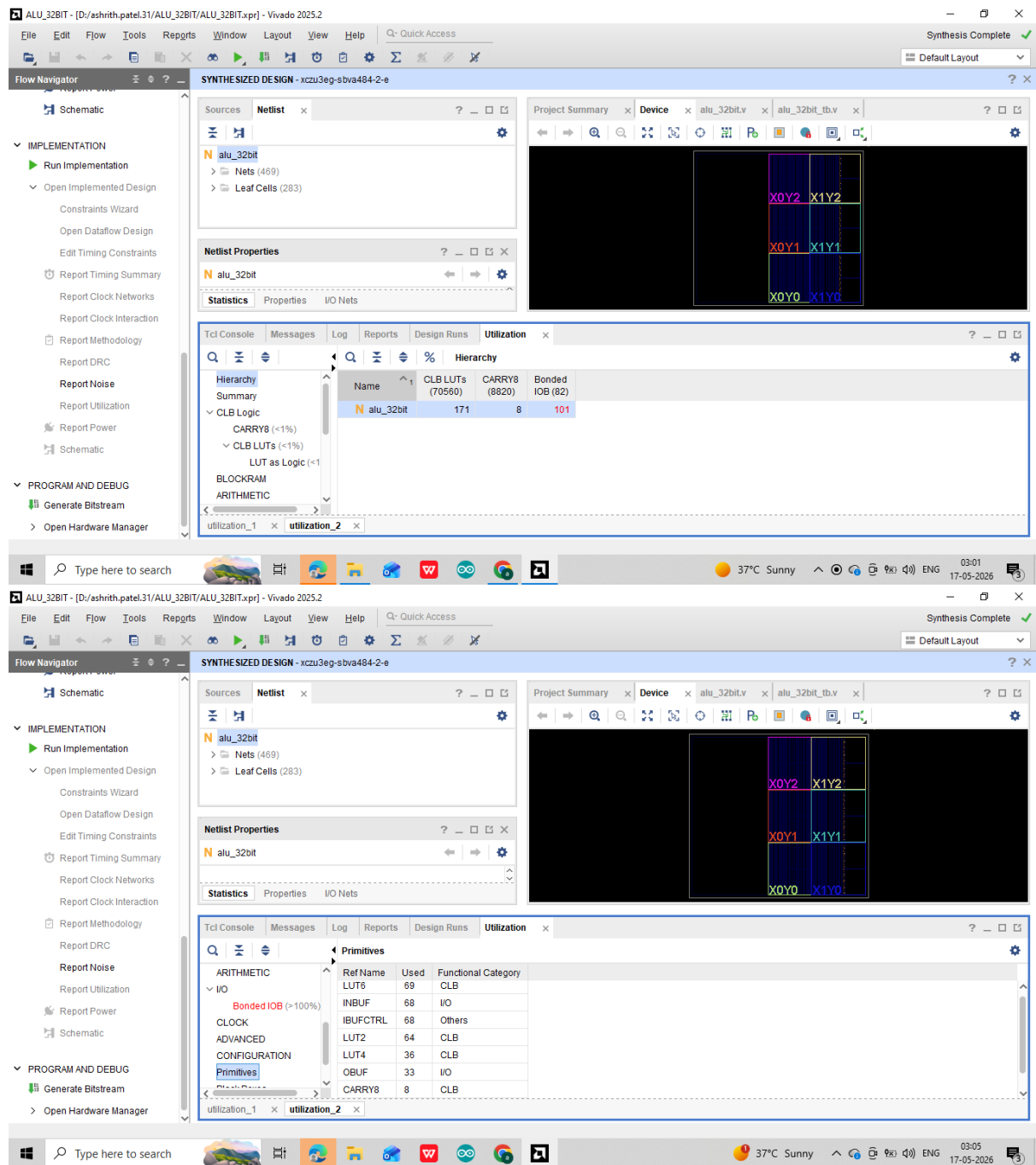


Figure 10.2: FPGA Resource Utilization Report

10.3 Timing Analysis

Timing analysis was performed to evaluate the operational reliability and timing performance of the designed ALU. The timing report generated by Vivado provided details regarding delay paths, slack values, and timing constraints.

The timing analysis confirmed that the design satisfied the required timing conditions without major timing violations, indicating stable and reliable hardware operation.

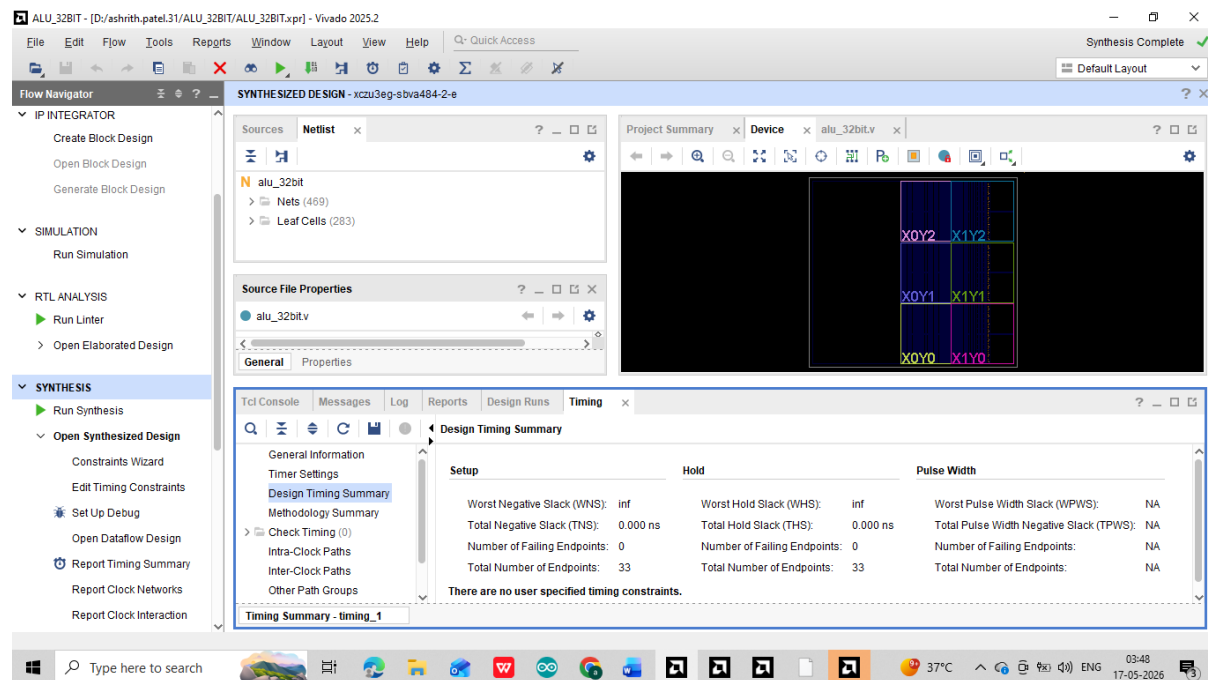


Figure 10.3: Timing Analysis Report

CHAPTER 11

ADVANTAGES

- High-speed arithmetic and logical processing
- Efficient FPGA resource utilization
- Supports multiple arithmetic and logical operations
- Behavioral modeling simplifies implementation
- Easy verification using simulation techniques
- Flexible and scalable architecture
- Suitable for FPGA and VLSI applications

CHAPTER 12

APPLICATIONS

- Microprocessors and microcontrollers
- Digital signal processing systems
- Embedded systems
- FPGA-based digital systems
- Communication systems
- Arithmetic processing units
- Computer architecture and processor design

CHAPTER 13

CONCLUSION

The 32-bit Arithmetic Logic Unit was successfully designed and verified using Verilog HDL and Vivado Design Suite. The ALU performed arithmetic and logical operations correctly according to the applied control signals.

Behavioral simulation, RTL analysis, and synthesis were completed successfully. The generated simulation waveforms, RTL schematic, and synthesis reports confirmed the correctness and efficiency of the design.

This project provided practical exposure to digital system design, RTL coding, simulation, synthesis flow, and FPGA implementation methodologies widely used in VLSI and embedded system applications.

CHAPTER 14

FUTURE SCOPE

- Implementation of pipelined ALU architecture
- Extension to 64-bit ALU design
- Low-power ALU optimization techniques
- FPGA hardware implementation on development boards
- Integration with processor architectures
- Support for multiplication and division operations

CHAPTER 15

REFERENCES

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CHAPTER 16

APPENDIX

The appendix section contains additional project-related information including complete Verilog source code, testbench code, additional simulation screenshots, synthesis reports, and waveform outputs used during verification of the 32-bit ALU design.

Additional screenshots and implementation details generated during Vivado simulation and synthesis processes are included for reference and documentation purposes.